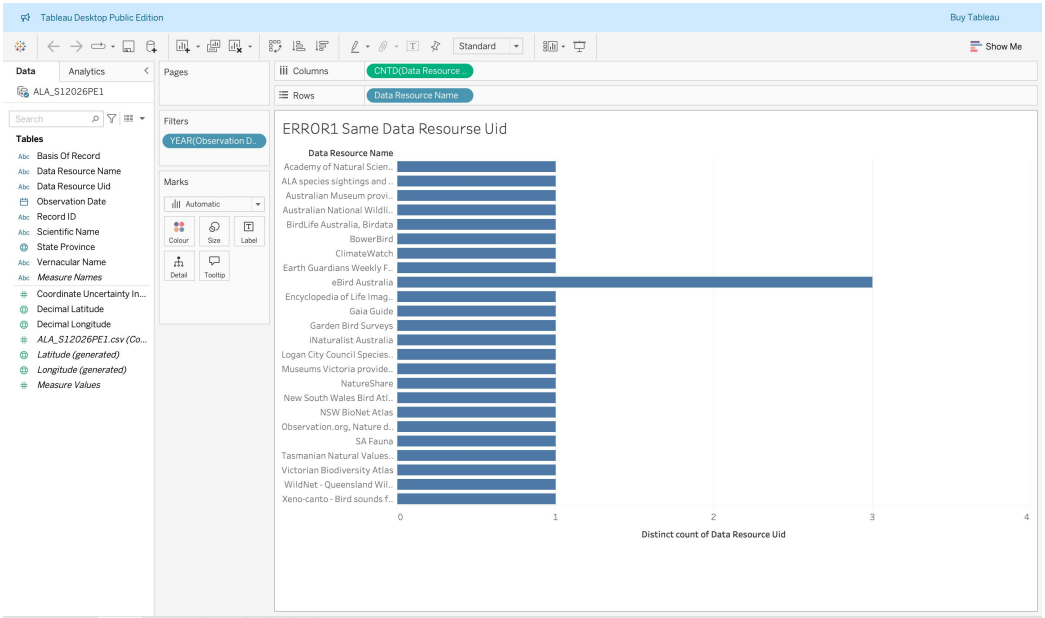
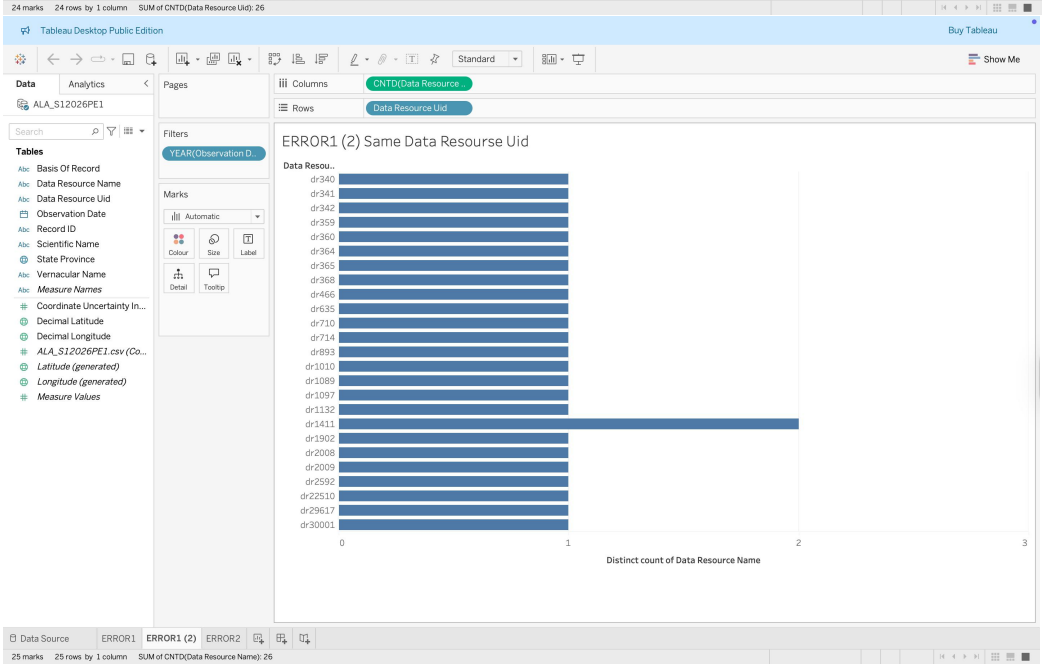
Name: Keshu Zhang, Student ID: 36436763, Tutor:Angel Das, Applied 10.

Swift Parrot Observation Data Analysis

1. Data Loading, Checking and Cleaning

Irregularity 1: Data Resource Mapping Conflict





- **Description & Discovery**:
  The first data irregularity is a logical conflict between the dataResourceUid and the dataResourceName. Under normal circumstances, these fields should have a strict one-to-one mapping. However, by cross-referencing these fields in Tableau, I discovered that 'eBird Australia' was incorrectly assigned multiple UIDs (dr2009, dr1411), and dr1411 was shared by two different resources. This confusion compromises the accuracy of aggregations based on data sources.
- **Resolution & Justification**:
  I cleaned the raw dataset in Excel by uniformly overwriting all 'eBird Australia' records with the correct dr2009 and correcting the name for dr1411 records to 'iNaturalist Australia'. This source-level cleaning ensures unique identification and prevents data fragmentation during analysis.

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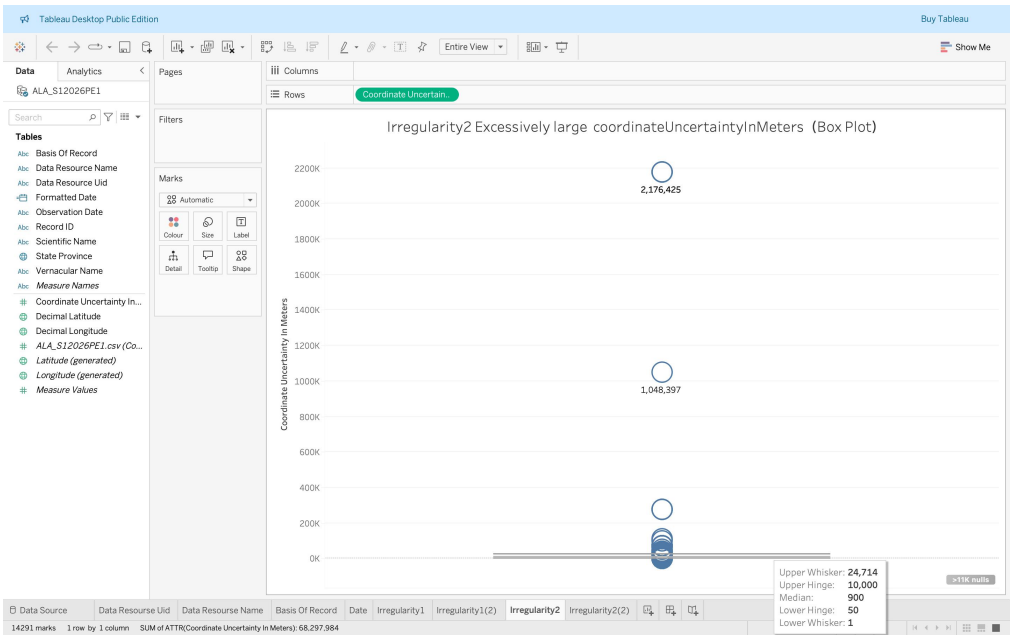
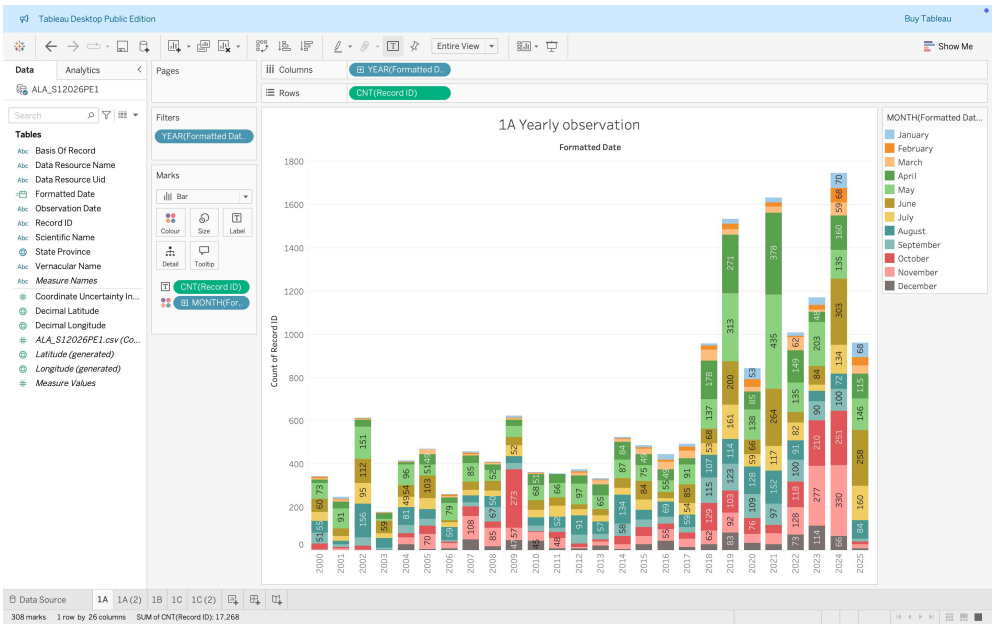
Irregularity 2: Extreme Coordinate Uncertainty



- **Description & Discovery**:
  I found few unreasonable extreme value in the coordinateUncertaintyInMeters field. This outlier severely exceeds normal error margins, causing the box plot to compress to the extent that the interquartile range is barely visible.
- **Resolution & Justification**:
  I investigated the field in Excel and chose to remove this extreme outlier. A spatial error of over 1,000 kilometers makes the record geographically meaningless. Discarding this invalid record is essential to maintain the rigour of subsequent spatial visualisations.

2. Data Exploration and Presentation

Q1A: Temporal Distribution of Observations



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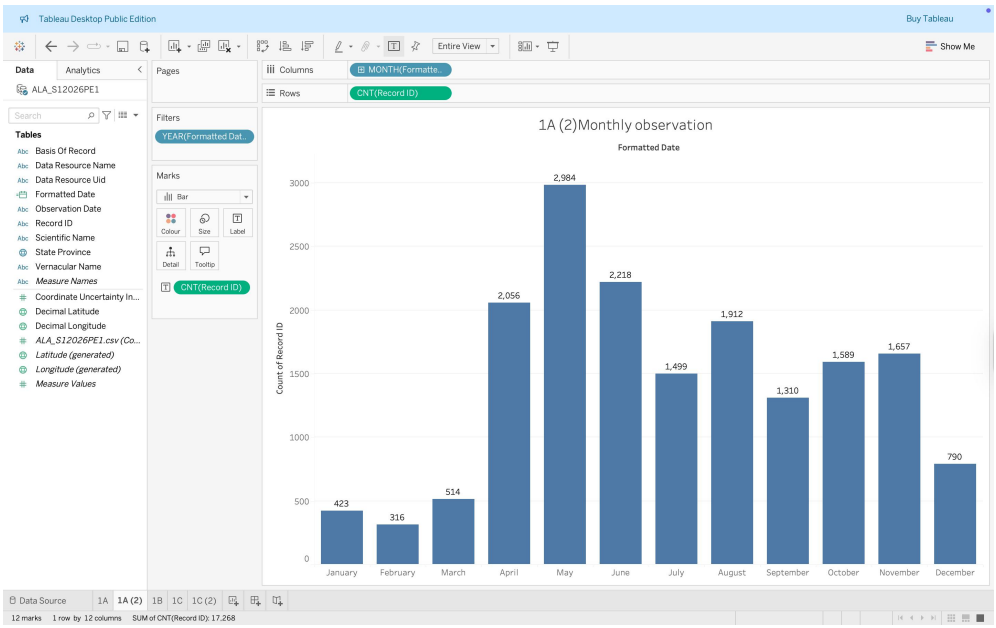
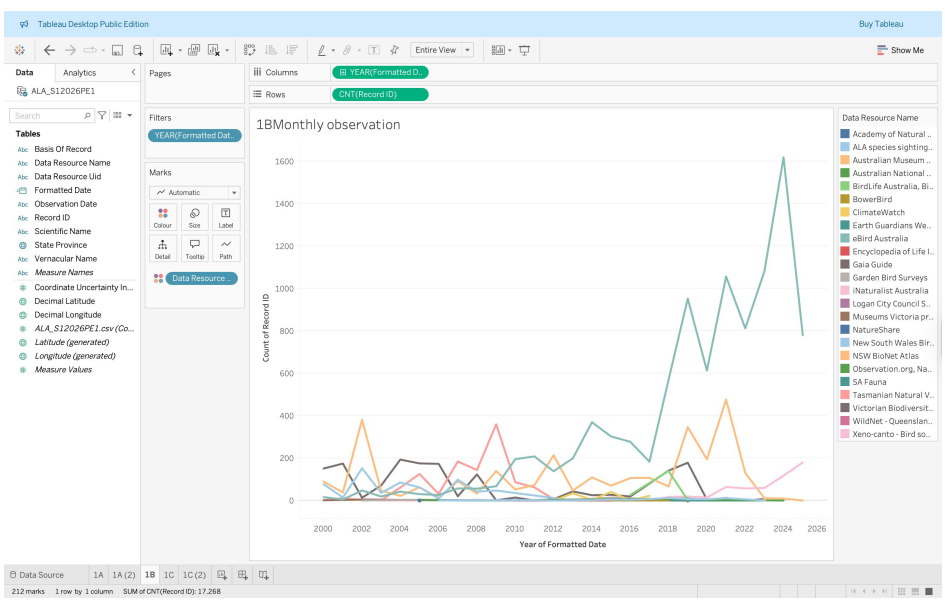


- **Visualisation Description & Justification**:
  I chose a line chart for the annual trend as it effectively highlights continuous temporal changes. For the monthly distribution, data was aggregated by month to reveal cyclical behaviours. Position on a common scale (y-axis) encodes the quantitative measure (Number of Records) for high interpretation accuracy. The x-axis position encodes the temporal data (Year/Month).
- **Insight 1 (Annual Trend)**:
  Observations generally increased since 2000, followed by significant drops in 2020, 2022, and 2023. This fluctuation is likely linked to extreme weather events like bushfires and habitat loss during those years.
- **Insight 2 (Monthly Trend)**:
  Records show significant seasonal variation, peaking around May and decreasing in January. This pattern reflects the seasonal movement of the species as they migrate from Tasmania to the mainland.

Q1B: Contribution of Data Resources Over Time



- **Visualisation Description & Justification**:
  A stacked bar chart was chosen to display both the total observation volume per year and the proportional contribution of each data resource.Length encodes quantitative observation counts. Color (Hue) differentiates the categorical dataResourceName, as it distinguishes platforms without implying a specific order.
- **Insight**:
  'eBird Australia' has experienced explosive growth over the past decade, becoming the dominant data source. Conversely, the 'Tasmanian Natural Values Atlas' records decreased significantly after 2012.

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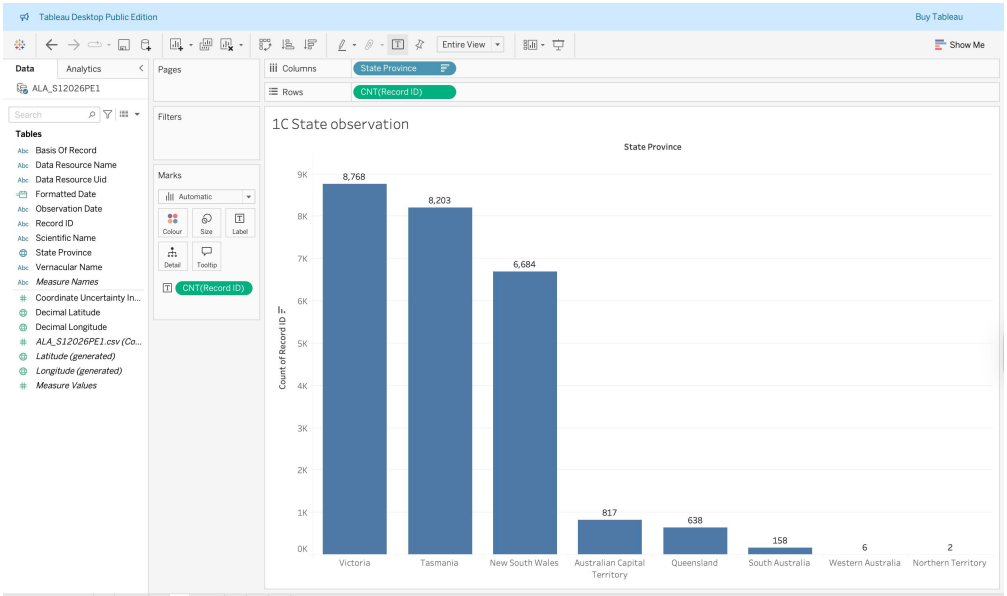
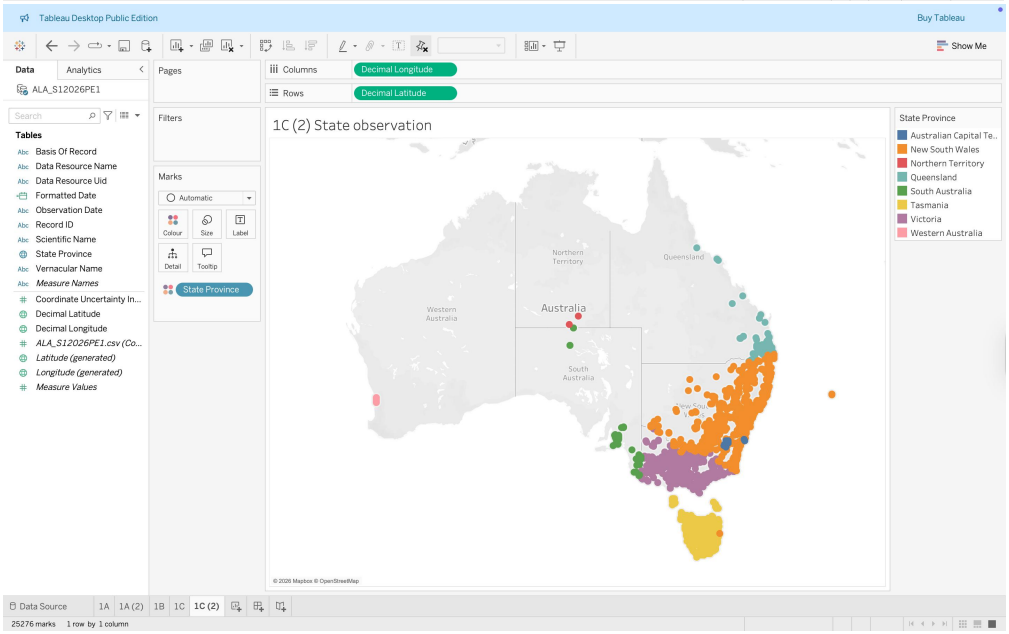
3. Geographical Distribution and Migration Reflection

Q1C: Geographical Distribution of Spotted Birds





- **Visualisation Description & Justification**:
  I used a dual-chart approach: a proportional symbol map for intuitive geographic context and a sorted bar chart for precise, rank-based quantitative comparisons.On the map, 2D Position denotes geographic location, and Size (Area) encodes the number of observations. In the bar chart, Length encodes record counts to make rankings instantly recognisable.
- **Insight**:
  Observations are heavily concentrated in south-eastern Australia, primarily in Tasmania, Victoria, and New South Wales. This distribution aligns with the core migratory range and habitat preferences of the species.

Q2: Reflection on Seasonal Migration

- **Answer**: No.
- **Justification:**To conclusively demonstrate migration, a visualisation must cross-reference 'Month/Season', 'Geographical Location', and 'Data Resource' in a single view. The generated charts are independent; 1A shows only temporal trends while 1C shows only spatial distribution. This information disconnect prevents us from deducing the dynamic movement of the species between regions across seasons.

Declaration of Generative AI Assistance

In the preparation of this paper, generative AI technology (Google Gemini) was utilized to assist with professional language refinement, structural polishing, and searching relevant knowledge.
Prompt:
Refine and enrich the content of my report, ensuring that it fully complies with academic professional standards and is within five pages.